

AI ENGINEER INTERNSHIP · TECHNICAL CASE STUDY

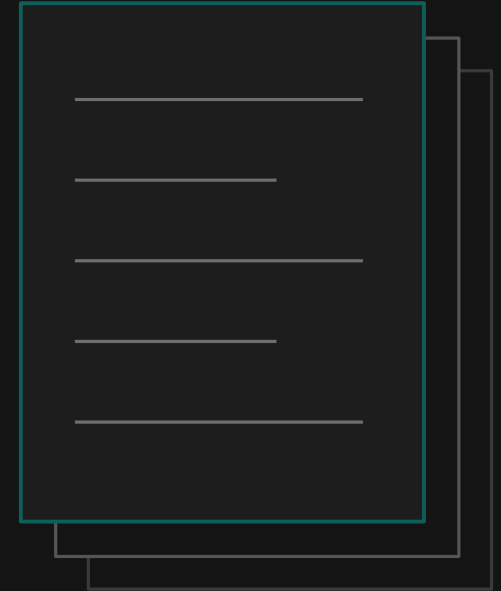
Intelligent Document Extraction, Validation & API Platform

OCR-driven extraction, deterministic financial validation,
and a deployed dashboard + REST API for financial documents.

Presented by

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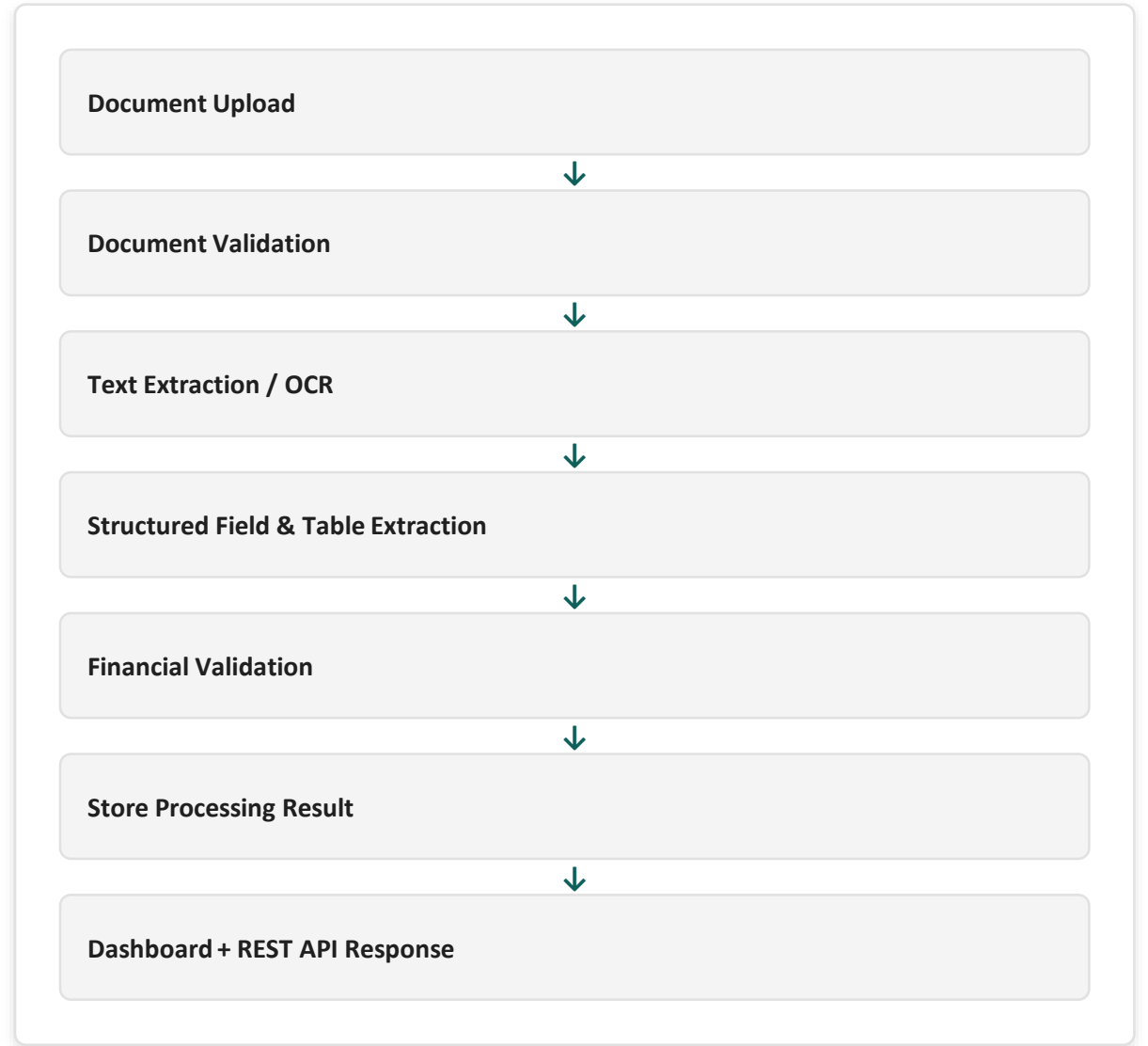
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Problem Statement

Financial-services teams frequently receive invoices and financial statements in native or scanned PDF, JPG and PNG formats. Different layouts, image quality, OCR errors and varying financial line items make manual extraction slow and inconsistent.

The goal is to build a reusable Document Intelligence platform that extracts meaningful information, validates financial relationships, and makes processed results available through a deployed dashboard and REST APIs.



SCOPE

Project Objectives & Scope

OBJECTIVES

- Extract meaningful information from financial documents
- Support PDF, JPG, JPEG and PNG inputs
- Support native PDFs and scanned / image-based documents
- Extract structured fields and tables / line items
- Preserve missing values as null
- Perform financial reconciliation checks
- Persist processed results in PostgreSQL
- Expose results through REST APIs
- Provide a web dashboard

SUPPORTED DOCUMENT CATEGORIES

01

Invoice

Line items, tax & totals

02

Balance Sheet

Assets vs. capital & liabilities

03

Profit & Loss

Income vs. expenditure

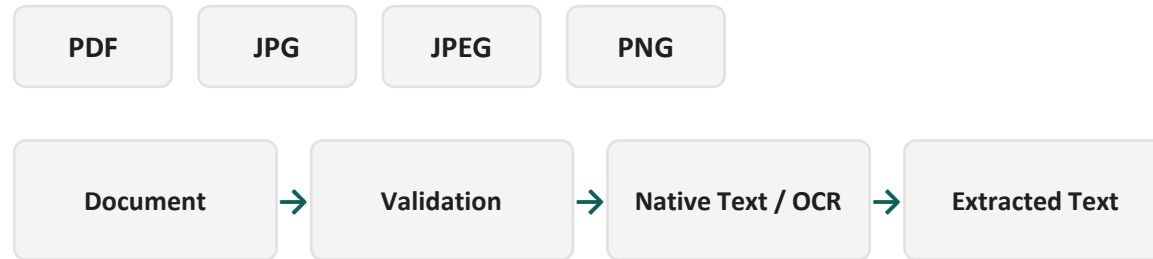
04

Cash Flow Statement

Operating, investing, financing

Document Processing & OCR

SUPPORTED INPUT



NATIVE PDF

PyMuPDF text extraction

SCANNED PDF / IMAGE

Image processing → Tesseract OCR → extracted text

VALIDATION

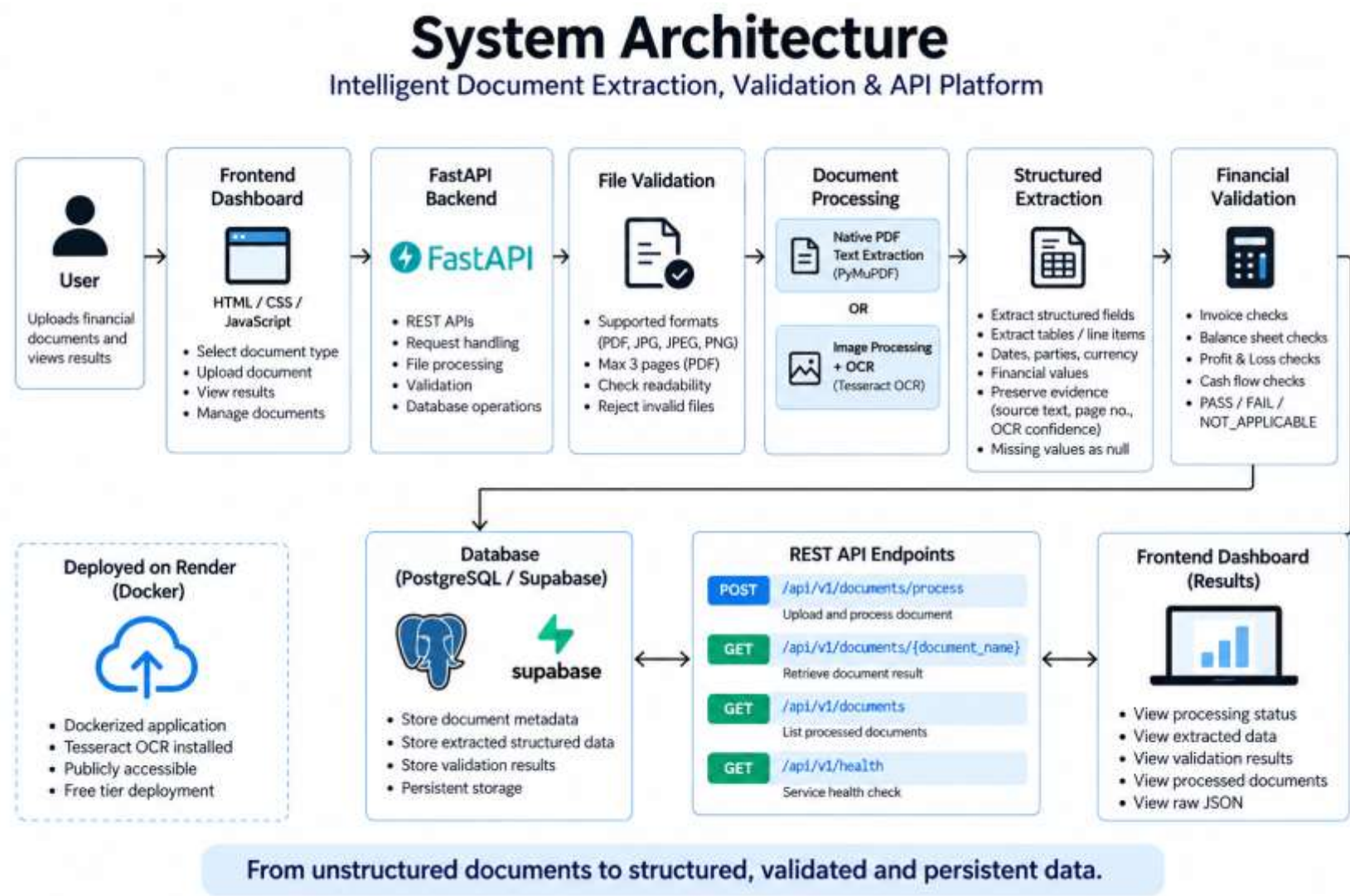
- Supported file type
- File integrity
- Maximum 3 PDF pages
- Readability check
- Invalid / corrupted files rejected

OCR & EXTRACTION METADATA

- Page-level text
- OCR words
- OCR confidence
- Readability status

Every extracted document carries the evidence used to derive it — not just the final structured values.

System Architecture



Structured Information Extraction

The platform converts unstructured document content into structured information ready for validation and storage.

EXTRACTS

- Document type
- Dates & comparative periods
- Parties & currency
- Financial values & totals
- Tables / line items

EVIDENCE CAPTURED

- Source text
- Page number
- OCR confidence, where available

Raw Document



Extracted Text



Structured JSON

DESIGN PRINCIPLE

“Missing information is returned as null, rather than being guessed or invented.”

Invoice Extraction & Validation

Document Result

Invoice

Processing Summary

Document Name

batch2-0499.jpg

Document Type

Invoice

Processing Status

PASS

File Type

image/jpeg

Pages

1

OCR Used

Yes

Financial Validation

PASS

Reconciliation checks performed on extracted data

Invoice line item 1 reconciliation

NOT APPLICABLE

Quantity * unit price = line total

Operands

```
{
  "period": "invoice",
  "quantity": null,
  "line_total": 18,
  "unit_price": 8
}
```

CAPTURED FIELDS

- Invoice number, date & due date
- Seller / buyer
- Currency
- Line items — quantity, unit price, tax
- Subtotal, shipping & total

VALIDATION EXAMPLES

Quantity × Unit Price ≈ Line Total

Subtotal + Tax + Charges ≈ Total

PASS

FAIL

NOT APPLICABLE

Profit & Loss Validation

Document Result

Profit & Loss

Dashboard

Processing Summary

Document Name

Consolidated Profit & Loss 2018.pdf

Document Type

Profit & Loss

Processing Status

PASS

File Type

application/pdf

Pages

1

OCR Used

Yes

Financial Validation

PASS

Reconciliation checks performed on extracted data:

P&L total income reconciliation - 2018

PASS

Interest earned + Other Income = Total Income

Openable:

"period": "2018",

"other_income": 16064644,

"interest_earned": 332678417

Calculated: 1013644478

Reported: 1013444478

Variance: 0

P&L total expenditure reconciliation - 2018

PASS

Interest expended + operating expenses + provisions & contingencies = total expenditure

Openable:

"period": "2018",

"interest_expended": 433034890,

"operating_expenses": 228272236,

"provisions_and_contingencies": 164790605

Calculated: 827836068

Reported: 827836068

Variance: 0

VALIDATION CHECKS

Interest Earned + Other Income
≈ Total Income

Interest Expended + Operating Expenses
+ Provisions & Contingencies ≈ Total Expenditure

Total Income - Total Expenditure
≈ Net Profit

Minority interest and attributable profit are validated where applicable.

PASS

FAIL

NOT APPLICABLE

INTELLIGENT DOCUMENT EXTRACTION, VALIDATION & API PLATFORM

08

Financial Statement Validation

BALANCE SHEET

Document Result

Balance Sheet

Dashboard

Processing Summary

Document Name

Consolidated Balance Sheet 2018.pdf

Document Type

Balance Sheet

Processing Status

PASS

File Type

application/pdf

Pages

1

OCR Used

Yes

Financial Validation

PASS

Reconciliation checks performed on extracted data

Balance Sheet main reconciliation - 2018

PASS

total capital & liabilities = total assets

Operands:

{

"period": "2018",

"total_capital_and_liabilities": 11031851695

}

Calculated: 11031851695

Reported: 11031851695

Variance: 0

Total Assets ≈ Total Capital & Liabilities

CASH FLOW STATEMENT

Document Result

Cash Flow Statement

Dashboard

Processing Summary

Document Name

Consolidated Cash Flow Statement 2018.pdf

Document Type

Cash Flow Statement

Processing Status

PASS

File Type

application/pdf

Pages

2

OCR Used

Yes

Financial Validation

PASS

Reconciliation checks performed on extracted data

Cash Flow net increase reconciliation - 2018

PASS

operating + investing + financing + fx/translation = net increase in cash

Operands:

{

"period": "2018",

"fx_translation": 280675,

"financing_cash_flow": 57370688,

"investing_cash_flow": -8821879,

"operating_cash_flow": 57046764

}

Calculated: 737504366

Reported: 737504366

Operating + Investing + Financing + FX ≈ Net Increase in Cash
Opening Cash + Net Increase ≈ Closing Cash

Validation Engine

EACH CHECK RECORDS

- Check Name
- Formula
- Operands
- Calculated Value
- Reported Value
- Variance
- Status

STATUS VALUES

PASS

Reported value matches the calculated value within tolerance.

FAIL

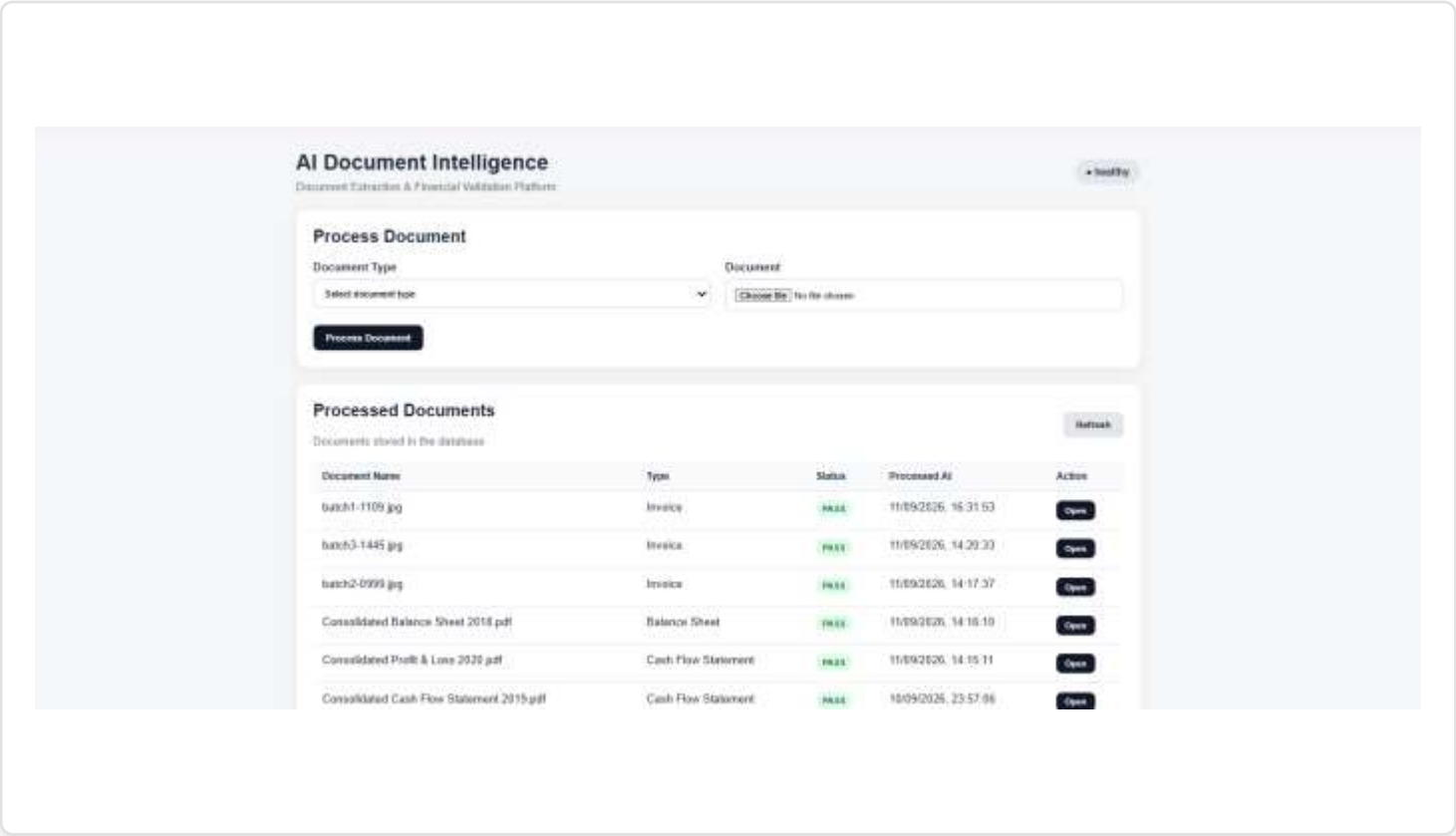
A material variance was found between calculated and reported values.

NOT APPLICABLE

Required inputs are unavailable, so no result is assumed.

If a required value is missing, the system does not assume a value — the result becomes NOT_APPLICABLE.

Frontend Dashboard



FEATURES

- Select document type
- Upload document
- View processing status
- View processed documents
- Open document result
- View extracted fields
- View tables / line items
- View validation results
- View processing metadata
- View raw JSON

Select Type → Upload → Process → View Result

REST API Platform

POST `/api/v1/documents/process`

Upload and process a document.

GET `/api/v1/documents/{document_name}`

Retrieve the latest processed result.

GET `/api/v1/documents`

List processed documents.

GET `/api/v1/health`

Service health check.

Swagger / OpenAPI: </docs>

WHAT THE API EXPOSES

The REST API exposes document processing, retrieval, listing and health-check functionality — the same operations the dashboard uses, available programmatically.

Database, Deployment & Testing

DATABASE

PostgreSQL / Supabase

- Document metadata
- Processing status
- Extracted structured data
- Validation results
- Processing metadata
- Timestamps

DEPLOYMENT

Render · Docker

- Dockerized application
- Tesseract OCR installed in deployment environment
- Publicly accessible service

TESTING

Functional coverage

- Health API tested
- Extraction tested
- Financial validation tested
- Invoice processing tested
- Scanned documents tested
- Unsupported file validation tested

Unsupported file → 400 Bad Request → **UNSUPPORTED_FILE_TYPE**

Technology Stack & Future Improvements

TECHNOLOGY STACK

Frontend	HTML, CSS, JavaScript
Backend	Python, FastAPI
PDF / Image	PyMuPDF, Pillow
OCR	Tesseract OCR
Validation	Python + deterministic financial rules
Data Validation	Pydantic
Database	PostgreSQL / Supabase
Deployment	Docker + Render
Version Control	Git + GitHub

FUTURE IMPROVEMENTS

- Better table and layout understanding
- Confidence thresholds and human review
- Additional document categories
- Authentication and role-based access
- Asynchronous processing
- Production monitoring and tracing
- Specialized document / vision models

LIVE DEMO & CONCLUSION

LIVE APPLICATION

ai-document-intelligence-g2cn.onrender.com

SWAGGER

ai-document-intelligence-g2cn.onrender.com/docs

GITHUB

github.com/shivanikushwaha131124/AI-Document-Intelligence

CONCLUSION

The platform provides an end-to-end workflow for transforming financial documents into structured and validated information.

Thank You

Document



OCR / Extraction



Structured Data



Financial Validation



Persistent Storage



REST API + Dashboard